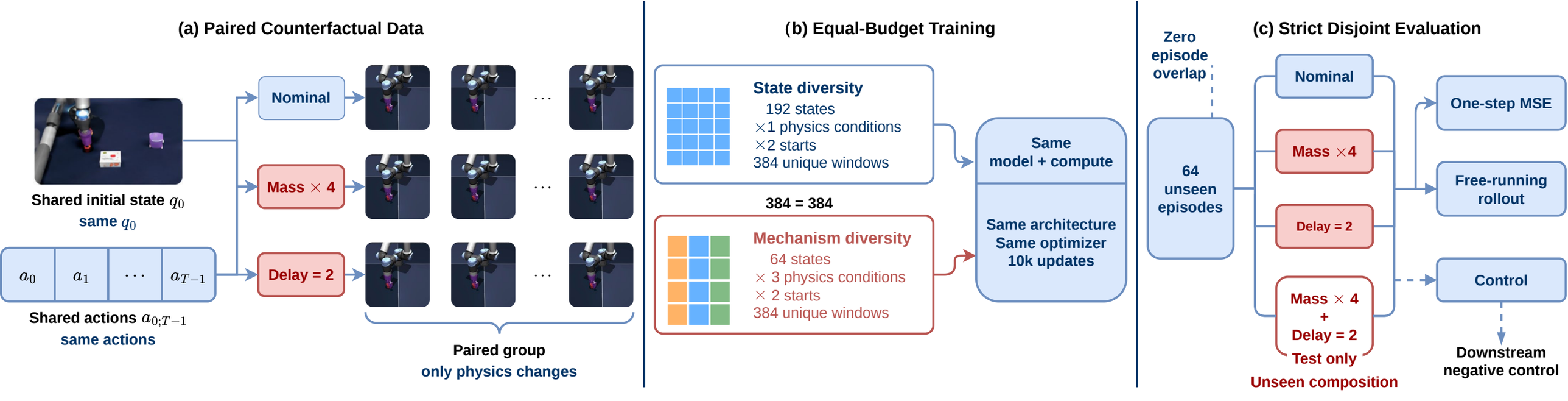

(a) Paired Counterfactual Data
Shared initial state q_0
same q_0
a_0
a_1
$\cdots$
a_{T-1}
Shared actions $a_{0;T-1}$
same actions
Nominal
Mass × 4
Delay = 2
Paired group
only physics changes
(b) Equal-Budget Training
State diversity
192 states
×1 physics conditions
×2 starts
384 unique windows
384 = 384
Mechanism diversity
64 states
× 3 physics conditions
× 2 starts
384 unique windows
Same
model + compute
Same architecture
Same optimizer
10k updates
(c) Strict Disjoint Evaluation
Zero
episode
overlap
64
unseen
episodes
Nominal
Mass ×4
Delay = 2
Mass × 4
+
Delay = 2
Test only
Unseen composition
One-step MSE
Free-running
rollout
Control
Downstream
negative control